

PROTOCOL FOR DIFFERENTIATION OF BLOOD-BRAIN BARRIER ENDOTHELIAL CELLS FROM HUMAN PLURIPOTENT STEM CELLS

Original protocol written by Ethan Lippmann 11/11/11
Updated by Hannah Wilson 4/30/14
Updated by Ethan Lippmann 1/25/16
Updated by Ethan Lippmann 8/1/16 to reflect new medium compositions
Updated by Emma Hollmann 10/21/16 to reflect use of UM
Updated by Emma Hollmann 6/16/17 to reflect use of E6
Updated by Emma Neal 4/12/19 to reflect use of B27

MATERIALS

I. Plasticware

FISHER

Corning Tissue Culture Plates (6- or 12-well, 3513 or 3516)
500 mL filter-top bottles (S2GPT05RE)

II. Reagents

Thermo Fisher Scientific

Human endothelial serum-free medium (hESFM)	11111	500 mL
Versene	15040-066	100 mL
B-27 Supplement (50X), serum free	17504-044	10 mL
Essential 8 Medium*	A1517001	500 mL
Essential 6 Medium*	A1516401	500 mL

****If desired, E8 and E6 may be purchased commercially rather than prepared in-house.***

Peprtech

Human basic fibroblast growth factor	100-18B	100 ug/mL
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Fisher

Human Holo-Transferrin, CF**	2914-HT-001G	1 g
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Sigma

All-trans retinoic acid	R2625	50 mg
Human insulin solution**	I9278	5 mL

*****If purchasing E8 and E6 commercially, human holo-transferrin and human insulin solution are not needed.***

REAGENT/MEDIUM PREPARATION:

E4 (prepared according to E4 large scale basal media production protocol)

Large batch previously prepared and stored at -80°C. If the preparation is not urgent, remove a bottle of E4 from the -80°C and place it in the fridge overnight, which will allow the bottle slowly thaw. However, if preparation is desired for the same day, remove a bottle of E4 from the -80°C, place it on the countertop for ~20 min, then place it in the 37°C water bath until it thaws completely (~1-2 h). Make sure to follow these steps precisely, as premature addition of a frozen bottle of E4 to a water bath can result in rupture of the plastic bottle.

Insulin

Pre-made solution provided by Sigma (catalog #I9278) that needs no additional preparation. Bottles are stored at 4°C.

Transferrin

Comes as a powder from R&D Systems (Human Holo-Transferrin, CF; catalog #2914-HT-001G). Add 50 mg of transferrin to 5 mL of phosphate-buffered saline (PBS), aliquot at 500 uL/vial, and store at -80°C. This mixture does not need to be sterile-filtered.

E6 media (prepared according to E6 and E8 media preparation protocol)

Dispense the thawed bottle of E4 into a bottletop filter attached to a 500 mL glass bottle. Add 100 uL of insulin solution and 500 uL of transferrin solution. Vacuum filter and store at 4°C. E6 media is stable indefinitely.

B27 Supplement

Thaw 10 mL bottle and mix thoroughly. Aliquot into sterile microcentrifuge tubes at 280 µL/tube and store at -20°C. Upon thawing, unused portions of an aliquot may be stored at 4°C for up to 1 week for further media preparation.

bFGF, 100 µg/mL (prepared according to E8 media protocol)

Thaw a 500 uL aliquot of bFGF and dilute 1:5000 in EC medium for a final concentration of 20 ng/mL as described below. Divide the remaining bFGF in 100 uL aliquots and re-freeze at -80 °C. These remaining aliquots can be thawed and used for EC medium but cannot be refrozen a second time.

Retinoic acid (RA)

Dilute 50 mg RA in 16.6 DMSO mL to create a stock solution of 10 mM and store 1 mL aliquots at -80 °C. To prepare working stocks, divide a 1 mL stock tube into 50 uL aliquots and store at -20 °C. Dilute working stocks 1:1000 in EC medium for a final concentration of 10 µM.

EC medium w/ 200X B27 + 20 ng/mL bFGF

For 50 mL: add 250 µL of B27 and 10 µL bFGF to 50 mL of hESFM.
Good for up to two weeks at 4°C.

EC medium w/ 200X B27

For 50 mL: add 250 µL of B27 to 50 mL of hESFM.

PROCEDURE:

Note: cells are seeded for differentiation in E8 medium according to the standardized single cell seeding protocol

1. BBB differentiation

Day 0-4

1. On day 0, aspirate E8 medium and add 2 mL of E6 per well.
2. Change medium every day using 2 mL of E6 per well.

2. BBB expansion

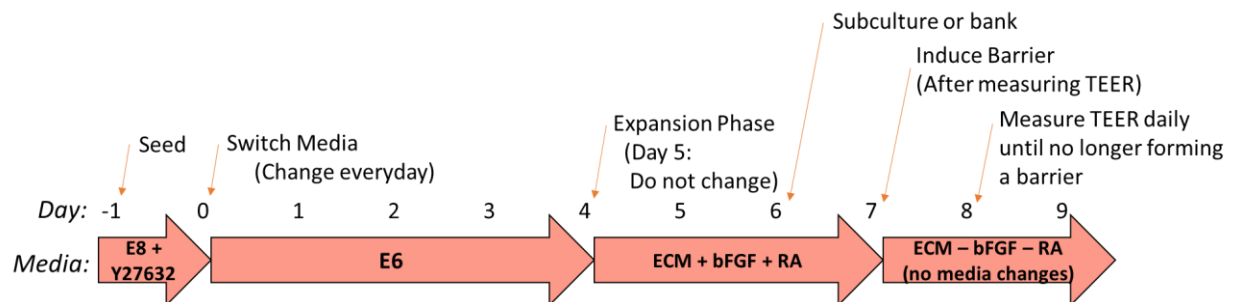
Day 4-6

1. At day 4 of E6 treatment, aspirate and add 2 mL of EC medium with bFGF and 10 μ M RA to each well.
2. Medium is NOT changed during expansion phase.

3. BBB subculturing

1. On day 6, subculture BBB onto plates and Transwell filters according to *BBB subculturing protocol*.

Schematic of E6 method for BBB differentiation



The above schematic was prepared by Amanda Bailey, VU undergrad.

We recommend the following antibodies to monitor BMEC differentiation [1]

Target antigen	Antibody species	Vendor	Clone or product number	Dilution
PECAM-1	Rabbit	Thermo Scientific	RB-10333P	1:10 (FC) 1:25 (ICC)
GLUT-1	Mouse	Thermo Scientific	SPM498	1:50 (ICC&FC)
Occludin	Mouse	Life Technologies	OC-3F10	1:100 (ICC) 1:50 (FC)
Claudin-5	Mouse	Life Technologies	4C3C2	1:100 (ICC) 1:50 (FC)
VE-Cadherin	Mouse	Santa Cruz Biotechnologies	F8	1:25 (ICC) 1:500 (FC)
E-cadherin	Goat	R&D Systems	AF648	1:100 (ICC&FC)
P-glycoprotein	Mouse	Life Technologies	F4	1:25 (ICC) 1:50 (FC)
Breast cancer resistance protein (BCRP)	Mouse	Millipore	5D3	1:25 (ICC) 1:50 (FC)
Multidrug resistance protein 1 (MRP1)	Mouse	Millipore	QCRL-1	1:100 (ICC) 1:50 (FC)
Glial fibrillary acidic protein (GFAP)	Rabbit	Dako	Z0334	1:500 (ICC)
β III tubulin	Rabbit	Sigma	T2200	1:1000 (ICC)
Nestin	Mouse	Millipore	10C2	1:500 (ICC)
α smooth muscle actin (SMA)	Mouse	American Research Products	1A4	1:100 (ICC)
Platelet-derived growth factor β (PDGFR β)	Rabbit	Cell Signaling	28E1	1:100 (ICC)

1. Lippmann, E. S.; Al-Ahmad, A.; Azarin, S. M.; Palecek, S. P.; Shusta, E. V. A retinoic acid-enhanced, multicellular human blood- brain barrier model derived from stem cell sources. *Sci. Rep.* 2014, 4, 4160.

Note: the VE-Cadherin antibody listed above is no longer appropriate.